WOOKYOUNG KIM, Ph.D.

Senior Researcher | Thermal Engineer

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PROFESSIONAL SUMMARY

Thermal engineer with 5+ years of R&D experience at Korea's national research institute (KIMM). Specialized in thermal management solutions for AI data centers, liquid hydrogen energy systems, and heat pump technology. Hands-on expertise spanning system design, experimental facility construction, high-pressure/cryogenic testing, and engineering software development (Python / FastAPI / React Native). Track record: 19 journal papers (11 SCI + 8 KCI), 18 domestic patents, 1 U.S. patent, 8 registered software programs, 5 technology transfers to industry.

CORE COMPETENCIES

Thermal Management	Energy Systems	System Design & Testing	Software & Analysis
Data center cooling (ILC/DLC)	Heat pump (vapor./ads.)	Heat exchanger (PCHE/S&T)	Python / FastAPI / React Native
Immersion / Jet impingement	Liquid hydrogen vaporizer	Thermal loop construction	2-phase HT measurement
Electronics thermal design	Low-GWP refrigerant R&D	High-pressure / cryogenic sys.	VLE measurement & EOS

EXPERIENCE

Senior Researcher | Korea Institute of Machinery and Materials (KIMM)

Jun 2021 - Present -- Korea's national research institute for machinery and materials

Data Center Thermal Management

- Jet-enhanced immersion cooling for next-gen high-heat-density servers [PI, 2025]
- Immersion cooling waste heat utilization and active thermal management [Lead, 2024-2028]
- Direct liquid cooling (DLC) system for data center power-consumption reduction [Lead, 2026-2029]

Cryogenic / Hydrogen Heat Exchangers

- PCHE design & testing for liquid-hydrogen vaporizers: cryogenic testing to -220°C / 100 MPa [Lead, 2021-2026]
- Anti-freezing PCHE design; CFD conjugate heat transfer analysis; freezing condition experimental mapping [Lead]
- Compact PCHE development for below -200°C, 100 MPa-class hydrogen supply system [Lead, 2022-2026]

Heat Pumps & Refrigerants

- 300°C-class high-temperature heat pump system for fossil fuel replacement [Participant, 2023-2028]
- Chemisorption heat pump with electrochemical compressor; experimental characterization [Participant, 2021-2025]
- Low-GWP refrigerant VLE measurement (R-32/R-125, R-1233ZD(E)) and equation-of-state development [Participant]

Engineering Software & Apps

- AI-based automated design tool for data center DLC cooling systems [PI, 2026]
- KIMMPROP: cross-platform iOS/Android thermophysical-property app (CoolProp/REFPROP via WASM) [PI, 2025]
- 8 registered engineering design programs (PCHE, heat-pump cycle, vapor chamber, etc.) [Lead]

Impact: 19 journal papers (11 SCI + 8 KCI) | 18 domestic patents + 1 U.S. patent | 8 registered software | 5 technology transfers

EDUCATION

B.S. Mechanical Engineering — **KAIST** — 2011 - 2015

M.S. Mechanical Engineering — **KAIST** — 2015 - 2017

Thesis: Effect of artificial cavities on the thermal performance of pulsating heat pipes

Advisor: Prof. Sung Jin Kim

Ph.D. Mechanical Engineering — **KAIST** — 2017 - 2021

Dissertation: Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes

Advisor: Prof. Sung Jin Kim

KEY ACHIEVEMENTS

19	18	1	8	5
Journal Papers (11 SCI + 8 KCI)	Domestic Patents	U.S. Patents	Registered Software	Tech Transfers

SELECTED PUBLICATIONS

(1) **W. Kim** et al., "Freezing Phenomenon in PCHE for Cryogenic LH2 Vaporizer," Appl. Therm. Eng. 273 (2025)

(2) **W. Kim** et al., "Freezing Condition of PCHE for LH2 Vaporizer," J. Hydrogen New Energy 35(2) (2024)

(3) **W. Kim** et al., "Falling Film Evaporation of R-1233ZD(E): Flow & Thermal Characteristics," Korean J. ACRE 36(1) (2024)

(4) **J.S. Kim, W. Kim** et al., "Pool boiling of ammonia outside enhanced tubes," Appl. Therm. Eng. 247 (2024)

(5) **H.S. Kim, W. Kim** et al., "Chemisorption heat pump performance under various conditions," Appl. Therm. Eng. (2024)

(6) **D.H. Kim, W. Kim** et al., "VLE of R-32/R-125: experiment and EOS verification," J. Mech. Sci. Technol. (2024)

(7) **J. Kim, W. Kim** et al., "Liquid behavior in falling-film evaporator distributor," Physics of Fluids 35 (2023)

(8) **W. Kim** and S.J. Kim, "Fundamental issues about pulsating heat pipes," J. Heat Transfer - ASME 143 (2021)

(9) **W. Kim** and S.J. Kim, "Flow behavior effect on pulsating heat pipes," Int. J. Heat Mass Transfer 149 (2020)

SELECTED PATENTS

Showing 8 of 18 domestic patents (Korean Intellectual Property Office)

- Immersion cooling device 2024
- Immersion cooling HVAC system and method 2022
- Heat exchanger with anti-freezing capability 2023
- Micro-channel reactor 2023
- Ternary refrigerant composition and heat pump system 2022
- Adsorption heat pump evaporator and system 2023
- Heat pipe integrated reactor for adsorption heat pump 2022
- Geothermal heat supply device and heating system 2023

REGISTERED SOFTWARE PROGRAMS

All 8 programs registered with Korea Copyright Commission

- 100 kg/hr hydrogen PCHE design program 2025
- Adsorption cooling simulation program 2025
- PCHE thickness design program 2024
- Vapor chamber performance analysis program 2024
- High-temperature heat pump cycle design program 2024
- Fin-tube heat exchanger design program 2024
- Heat exchanger HTC uncertainty analysis program 2023
- Lab-scale heat exchanger analysis program 2023

TECHNOLOGY TRANSFERS

5 transfers to industry partners

- Heat pipe and vapor chamber structural design for electronics cooling
- Micro-channel bonding and fabrication techniques for heat exchangers
- Low dew-point dehumidifier development technology
- Ultra-thin vapor chamber for next-generation semiconductor cooling
- Compact inverter compressor-based dress care system development

TECHNICAL SKILLS

Experimental: Thermal loop design & construction (1-/2-phase) · Low-GWP refrigerant systems · 2-phase flow & heat-transfer measurement · High-pressure testing (100 MPa) · Cryogenic systems (-220°C) · Flow visualization

Analytical & Computational: Thermal network modeling · Heat exchanger design (PCHE, S&T, PHE) · CFD (ANSYS FLUENT, COMSOL) · CAD (SOLIDWORKS, INVENTOR) · Machine learning & surrogate modeling (scikit-learn, PyTorch, LightGBM) · CoolProp/REFPROP

Software Development: Python · JavaScript/TypeScript · C/C++ · FastAPI · React / React Native · Git · Docker · Linux

PROFESSIONAL ACTIVITIES

- **Council Member**, Korean Society of Mechanical Engineers (KSME) [2026.04 - Present]
- **Committee Member - Thermal Management / Cooling**, SAREK Data Center Technology Committee [2026.03 - Present]
- **Regular Member**, Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK) [2022.04 - Present]
- **Regular Member**, Korean Society of Mechanical Engineers (KSME) [2021.05 - Present]